

Joule heating for SMA wire

1. Introduction

Previous documentation utilized an ideal model that accepted direct temperature signals without accounting for real-world thermal dynamics. By introducing the electro-thermal equations of the Joule heating system, this work models the real-world behavior of shape memory alloy (SMA) wires under practical heating and cooling conditions

2. The configuration of Joule heating system

System Operation and Control Strategy

Prior to presenting the governing electro-thermal equations, it is essential to outline the operational mechanics of the Joule heating system. The system utilizes a 12V DC power supply to heat the wire. Because the electrical power source operates as a binary switch (either fully on or completely off with no intermediate values), activation time serves as the primary mechanism for reaching a target temperature.

For example, assuming a baseline wire resistance of 1 Ω , the total available electrical power is calculated as:
 $P = V^2 / R = 12^2 / 1 = 144 \text{ W}$

To reach the target temperature, this 144W load must be applied across the wire for a calculated duration. Once the desired temperature is achieved, Pulse Width Modulation (PWM) is utilized to maintain it.

Pulse Width Modulation (PWM) and Duty Cycle

The primary control input for the PWM signal is the duty cycle, which dictates the percentage of time the power remains active within a single operating cycle. For instance, at an operating frequency of 1 Hz (one cycle per second), a 50% duty cycle means the 144W load is applied for exactly 0.5 seconds and turned off for the remaining 0.5 seconds. Because PWM can operate at higher frequencies, the power cycles rapidly, allowing precise temperature maintenance by adjusting this percentage.

- **Heating Phase:** To transition from an ambient temperature of 293 K to a target of 300 K, the controller applies a 100% duty cycle (continuous full power).
- **Maintenance Phase:** Once at 300 K, the duty cycle drops below 100% to balance heat loss and sustain the temperature. Sustaining a 100% duty cycle would cause overheating, while dropping to 0% would let the wire cool back down to room temperature.

The fundamental thermodynamic behavior of the system dictates that the net rate of temperature change is a function of both the electrical energy input and environmental thermal interactions. Specifically, the rate of temperature change over time is defined as the electrical power multiplied by the reciprocal of the thermal mass, minus the convective heat loss multiplied by the reciprocal of the thermal mass.

Mathematically, this relationship is expressed as:

$$\frac{dT}{dt}_{\text{net}} = \left(\frac{V^2}{R \cdot mC_p} \right) - \left(\frac{hA_s(T - T_{\infty})}{mC_p} \right)$$

Circuit Regulation and Safety

The overall system is regulated by a PID controller configured with a saturation limit of (0, 1) to restrict the duty cycle output. Additionally, a MOSFET is integrated into the circuit for overcurrent protection. Introducing the core mathematical model is necessary to formally establish how this Joule heating process behaves.

The power supply voltage (V)	12 V
The wire resistance (R)	1.08 Ohm
The mass of the wire (m)	.25 gram
The specific heat capacity (Cp)	400 J/Kg.K
The convective heat transfer coefficient (h)	515 W/m ² .K
The surface area (As)	1.5708 * 10 ⁻⁴ m ²
The room temperature (T _∞)	293 K

Table (1)The electro-thermal parameters

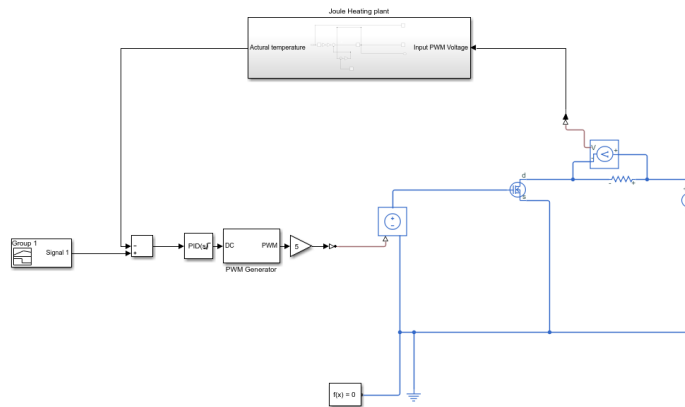


Figure (1) The block diagram of the system.

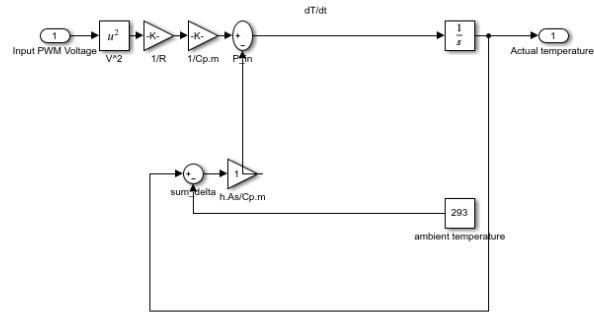


Figure (2) The joule heating system

3. The control of Joule heating

This section evaluates the transient performance of the Joule heating system during distinct heating and cooling phases. Furthermore, it analyzes how variations in the convective heat transfer coefficient influence the thermal response rate and system efficiency.

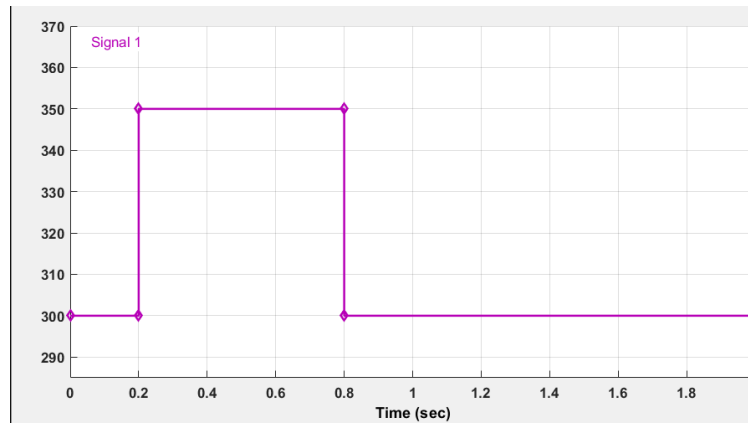


Figure (4) The input temperature of the Joule heating

The commanded temperature trajectory initializes at 0 seconds, where the system is required to step from an ambient temperature of 293 K up to a baseline of 300 K. At the 0.2-second mark, a second step command forces a rapid rise to a peak temperature of 350 K. Finally, at 0.8 seconds, the target signal drops back down, instructing the system to cool down and stabilize at 300 K.

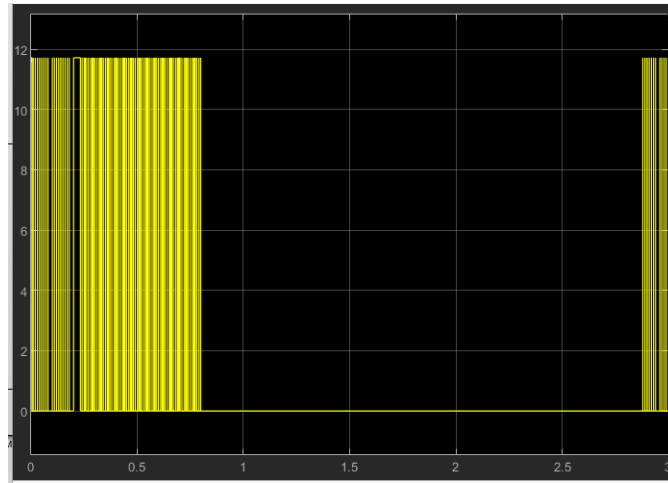


Figure (5) The PWM input Voltage

The duty cycle profile reflects the energy demands of the commanded trajectory. To achieve the step increase up to 350 K, the PWM signal increases in density to maximize electrical power delivery. Once the cooling phase begins, the applied voltage drops immediately to zero, allowing the wire to shed heat naturally and fall toward 300 K. After reaching this lower target, the signal density increases once again to balance ongoing convective heat loss and stabilize the temperature.

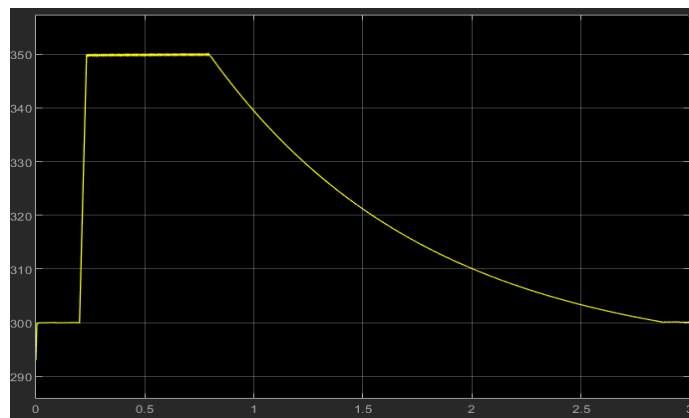


Figure (6) The output temperature

The system exhibits a rapid initial thermal response, with the wire temperature rising to 300 K within the first 0.05 seconds. This rapid rise corresponds to the high-density PWM signal required to drive the step transition. Once the target temperature is achieved, the PWM duty cycle decreases to a lower density, supplying just enough power to offset convective losses and maintain the 300 K baseline. To meet the subsequent command to reach 350 K, the

controller maximizes power delivery, causing the PWM signal to become dense once again; it remains saturated at a high density due to the high energy required to sustain this elevated temperature.

When the target drops back down to 300 K, a noticeable two-second cooling delay occurs before the wire temperature descends to the baseline, at which point the PWM signal returns to maintain stability. To explore methods for reducing this delay, a subsequent experiment triples the convective heat transfer coefficient (h). In a physical setup, this parameter adjustment represents switching to a more aggressive cooling method.

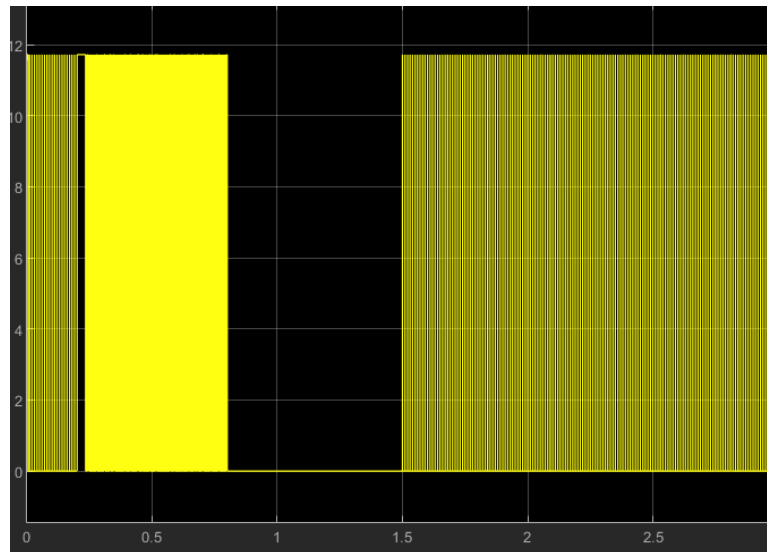


Figure (7) The second input PWM voltage

With the increased convective heat transfer coefficient, the controller operates at a significantly higher duty cycle to overcome the heightened heat dissipation, requiring denser power delivery to reach and sustain the target temperatures. Consequently, this aggressive cooling method successfully reduces the cooling step delay from the initial two seconds down to 0.7 seconds

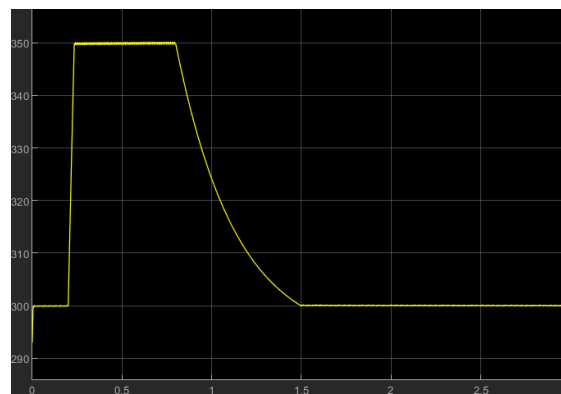


Figure (8) The output temperature of the second experiment

The resulting temperature trajectory directly mirrors the behavior of the preceding PWM input profile. The sudden termination of power delivery forces the system into a passive cooling phase. Once the wire temperature descends to the 300 K baseline, the controller automatically scales up the duty cycle to supply the necessary power to balance heat loss and maintain a steady-state temperature.

4. Integrating Joule heating to SMA bending wire system

Following the standalone validation of the Joule heating subsystem, the thermal model is integrated into the primary mechanical system to achieve a higher-fidelity simulation of real-world behavior. Prior to integration, the heat transfer equation must be modified to account for latent heat. This phase-transformation enthalpy acts as a thermal buffer, slowing down both the heating and cooling transients. The formulation incorporates the latent heat of transformation enthalpy ($\Delta H = 24,000 \text{ J/kg}$) alongside the rate of change of the martensite volume fraction ($\frac{d\xi}{dt}$)

$$\frac{dT}{dt} = \frac{1}{m \cdot C_n} \left[\frac{V^2}{R} - hA_s(T - T_\infty) + m\Delta H \frac{d\xi}{dt} \right]$$

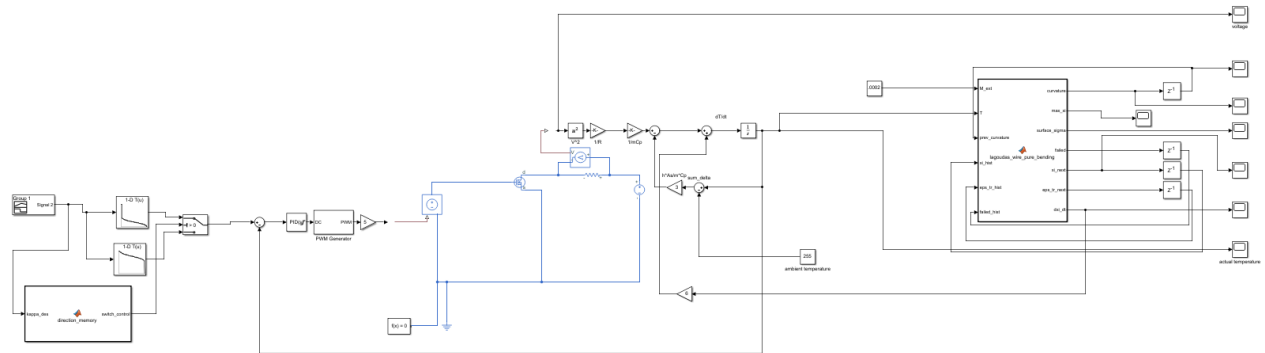


Figure (9) The Joule heating integrated SMA Bending wire

To maintain complete transparency in the system design and clearly illustrate all operational dynamics, the model components are kept fully exposed rather than being consolidated into hidden subsystems.

The control sequence initializes when the reference position signal is passed through a configured lookup table. This table maps the target position to its corresponding target temperature, which is then transmitted directly to the Joule heating assembly to initiate the wire heating cycle.

Additionally, the environmental parameters for this simulation run have been modified: the ambient temperature is set to a sub-zero environment of 255 K, and the latent heat of transformation detailed previously is fully incorporated into the thermal model

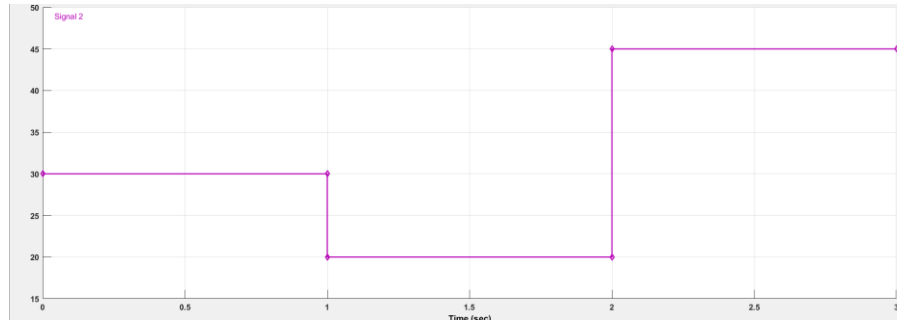


Figure (10) The input desired curvature trajectory signal



Figure (11) The input PWM voltage

Although visual resolution constraints within the plot limit the ability to distinguish local variations in PWM density, the structural transitions of the profile remain evident. Specifically, the onset of the cooling phase is clearly identifiable, accompanied by a distinct 0.6-second delay before thermal stabilization occurs.

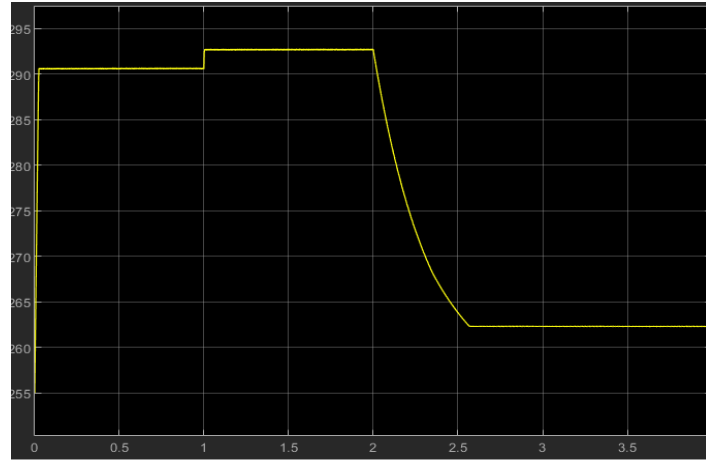


Figure (12) The applied temperature on the SMA wire

The response highlights the gradual cooling profile as the system transitions toward its final destination temperature of 263 K.

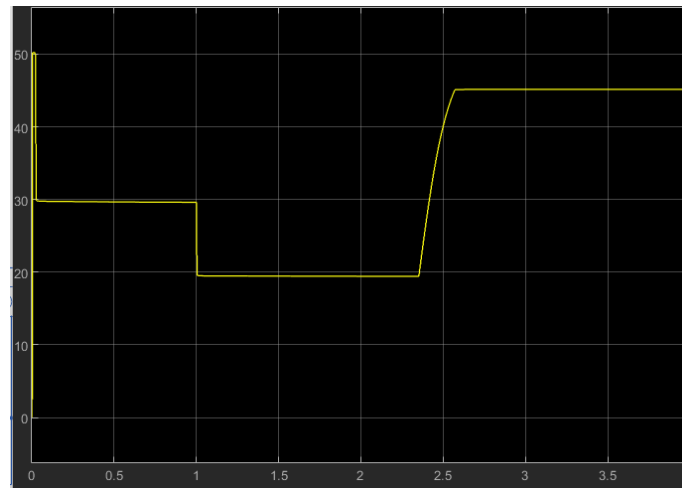


Figure (13) The output curvature

The mechanical bending load directly affects the system dynamics, forcing the wire to an initial curvature of 50 m^{-1} . The wire then transitions toward a 30 m^{-1} curvature, continues along its pathway down to 20 m^{-1} , and ultimately drives toward its final destination of 45 m^{-1} . Throughout this trajectory, phase transformations within the material introduce a distinct 0.6-second cooling delay alongside a 0.35-second dead-band zone.

5. The SMA wire safety and the balance of convection method and the applied voltage

While the initial system configuration operated at 12V, executing this high voltage in physical applications poses significant hardware risks. Overheating can cause permanent structural damage to the crystalline composition of the shape memory alloy, effectively destroying its shape memory effect. To mitigate this risk, the system must balance electrical input and thermal dissipation; if input power declines due to a lower applied voltage, the cooling rate must also be managed so it does not overwhelm the reduced electrical heating capacity.

Consequently, this section evaluates system performance under a safe operating potential of 3V coupled with a baseline convection coefficient ($h = 515 \text{ W/m}^2 \cdot K$). This configuration establishes that transient response speed cannot be prioritized at the expense of material integrity and wire safety.

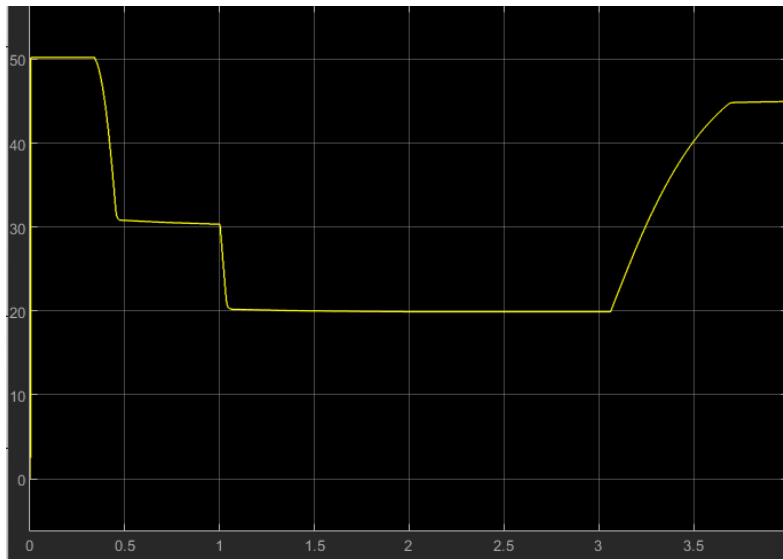


Figure (14) The output curvature

The low-voltage response exhibits a noticeable 0.5-second actuation delay during the initial step command. In physical applications, this delay can be effectively mitigated by pre-heating the wire to a baseline temperature just below its austenite start temperature (A_s), maintaining this primed state regardless of whether an active command signal is present.

With the heating delay addressed, the system focus shifts to optimizing the cooling profile. A proper cooling method must be carefully balanced to reduce the current 1.6-second cooling step delay without overpowering the limited 3V electrical capacity.

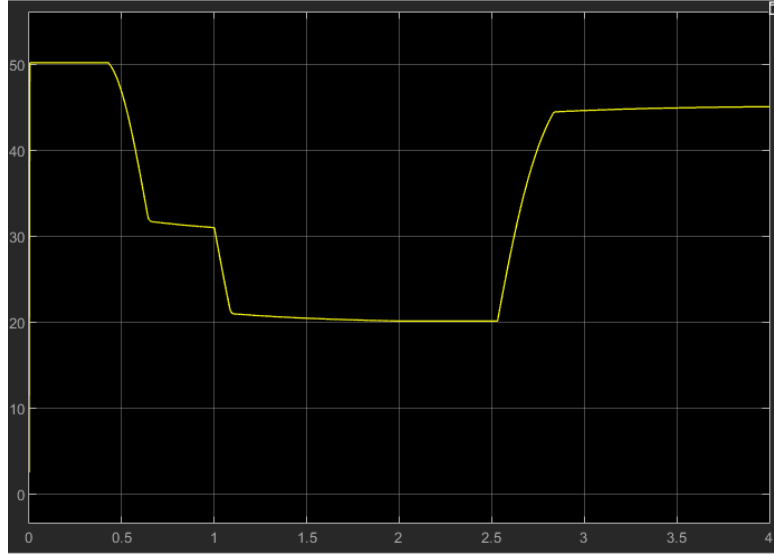


Figure (15) the output curvature with convection gain doubled

A comparative analysis of the system responses highlights a noticeable transient tracking error. The wire approaches its target trajectory at the 0.6-second mark but fails to achieve the commanded 30 m^{-1} curvature until a full 1.0 second has elapsed. This lag repeats during the subsequent 20 m^{-1} step transition.

This behavior is entirely a proportional-integral (PI) tuning artifact rather than a fundamental physical limitation of the system. Consequently, instead of adjusting the parameters of the inner-loop temperature PI controller, this tracking error will be resolved by implementing a hybrid feedforward control architecture in subsequent development stages.

6. The conclusion

This document successfully evaluated the operational mechanics of Pulse Width Modulation (PWM) regulation combined with an electro-thermal Joule heating model. The findings demonstrate that increasing the convective heat transfer coefficient—simulating a shift to a more aggressive physical cooling method—forces the controller to increase duty cycle density to reach and maintain target temperatures.

Transitioning from a standalone model to an integrated framework provided a high-fidelity representation of a real-world shape memory alloy (SMA) wire actuator. While the added latent heat dynamics showed a negligible impact during active heating phases, their buffering effect became highly pronounced during cooling transitions. However, this delay can be effectively suppressed by optimizing the environmental cooling method in the physical setup.

Furthermore, wire safety and material integrity remain paramount; exposing the SMA to high voltage for rapid actuation is a critical hardware risk that can permanently degrade its crystalline composition. Real-world applications must carefully balance the applied operating voltage with the selected cooling configuration.

Two critical anomalies remain unresolved: the management of varying mechanical bending loads and the transient tracking errors identified during closed-loop testing. To preserve the concise nature of this analysis, these remaining challenges will be comprehensively addressed in a standalone, subsequent documentation.

